Price1-Only Ruleout Threshold Robustness without Dynamic Sharing

BoardPharma

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1 Price1-Only Ruleout Definition and Common Regression Design

Within each saved cohort panel, observations are grouped by company and product (`BoardName` × `product`). Using all observed quarters from the earliest through the latest quarter in that cohort file, the price1-only ruleout score for company b, product p, and cohort c is

$$R_{bpc}^{\text{price1}} = \frac{\max(\texttt{price1}_{bpc})}{\min(\texttt{price1}_{bpc})}.$$

For threshold τ, every row belonging to that company–product history is removed from that cohort when $R_{bpc}^{\text{price1}} > \tau$. Equality is retained. The score uses original price levels and is calculated before the event-time restriction, upper-p95 winsorization, and log transformation. Ruleout = 0 is the unfiltered benchmark. The other panels set τ to 1000, 100, or 10.

Treatment status, control status, event timing, event cohort, and ATC3 sharing are read from the same saved cohort files and are not redefined by ruleout. Dynamic sharing is disabled in all four runs. After company–product histories are removed, a cohort with no remaining treated observations is not stacked. The retained sample then passes through the common regression procedure: the $[-4, 7]$ event window, upper-p95 winsorization of price1, the log transformation, fixed effects, company clustering, and `did_imputation` autosample selection.

All four runs use the same inputs and regression settings: the `drop_sentinel_yearquarter` roster, `ever_board=1`, `ieoc_ref=1`, `late=3`, new requirement, stay, and sharing-time rules, non-dynamic ATC3 sharing, `price1`, req2, the Not control group, company–product–cohort fixed effects, cohort–calendar-quarter fixed effects, company clustering, horizons 0–7, three pre-event coefficients, and no Kappa, other-event, or ATC3 fixed-effect controls.

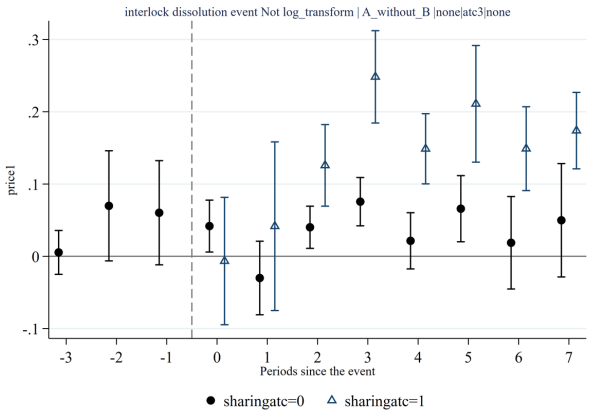
Table 1: Estimation sample across the six event–side regressions

Ruleout setting	Total observations	Reduction from ruleout = 0
0 (off)	583,584	0.00%
1000 (price1)	582,420	0.20%
100 (price1)	580,560	0.52%
10 (price1)	551,316	5.53%

Notes. Counts sum the final estimation observations reported by the six output files and therefore refer to stacked regression observations, not unique company–product records.

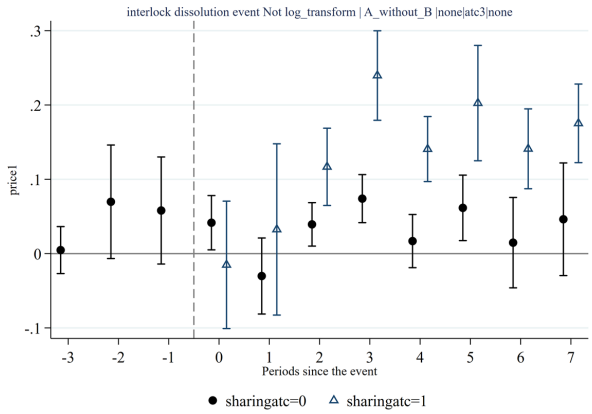
2 Interlock Dissolution — A-side Treatment

Ruleout = 0



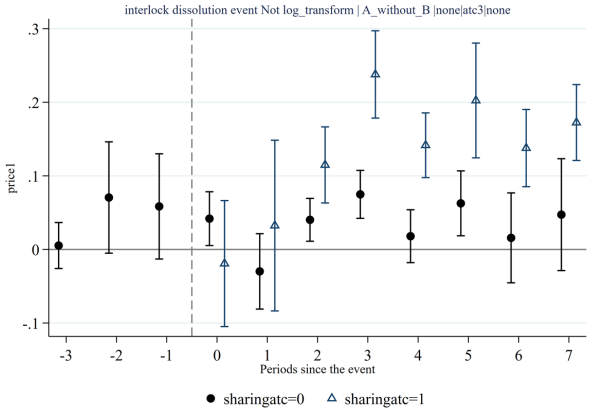
Estimation observations: 109,824

Ruleout = 1000



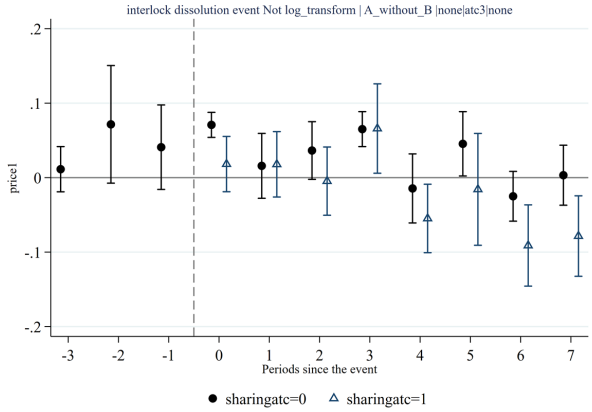
Estimation observations: 109,632

Ruleout = 100



Estimation observations: 109,236

Ruleout = 10

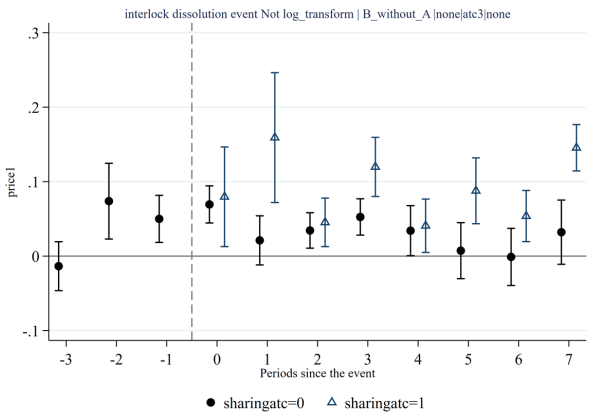


Estimation observations: 105,708

Notes. Black circles report estimates for `sharingatc=0`; hollow blue triangles report estimates for `sharingatc=1`. Error bars are 95% confidence intervals. The panels differ only in whether ruleout is off or in the price1-only threshold, and therefore in the resulting estimation sample. Dynamic sharing is disabled in all four runs.

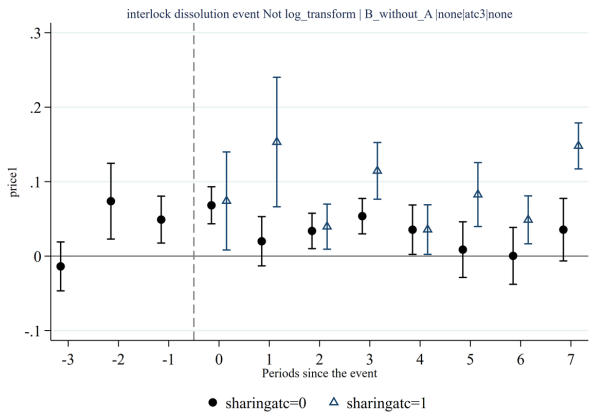
3 Interlock Dissolution — B-side Treatment

Ruleout = 0



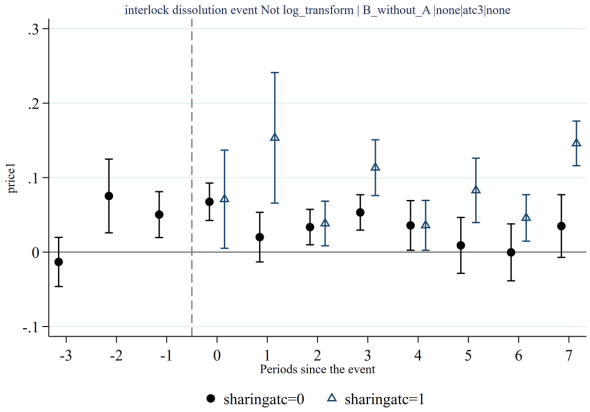
Estimation observations: 94,944

Ruleout = 1000



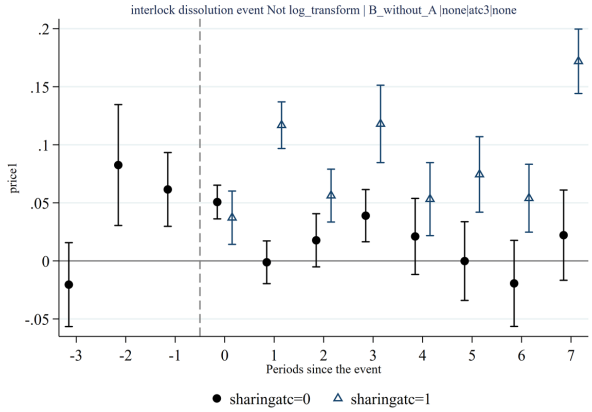
Estimation observations: 94,764

Ruleout = 100



Estimation observations: 94,440

Ruleout = 10

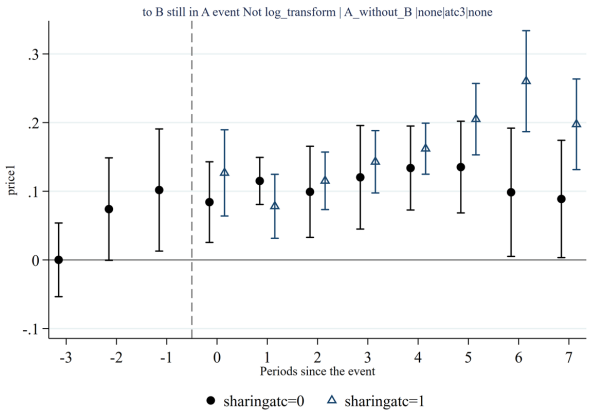


Estimation observations: 88,332

Notes. Black circles report estimates for `sharingatc=0`; hollow blue triangles report estimates for `sharingatc=1`. Error bars are 95% confidence intervals. The panels differ only in whether ruleout is off or in the price1-only threshold, and therefore in the resulting estimation sample. Dynamic sharing is disabled in all four runs.

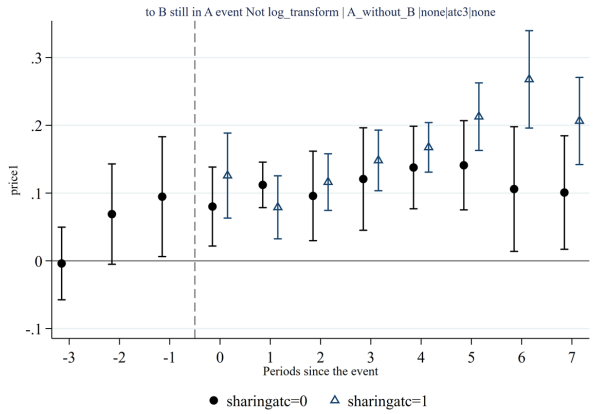
4 To B, Still in A — A-side Treatment

Ruleout = 0



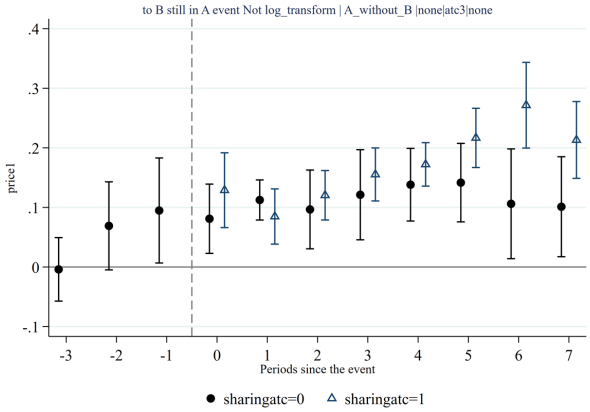
Estimation observations: 108,396

Ruleout = 1000



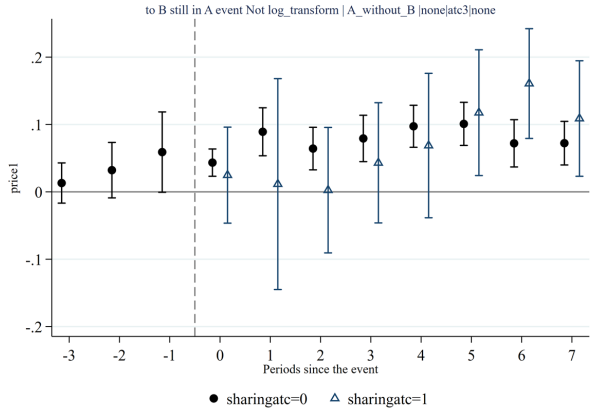
Estimation observations: 108,168

Ruleout = 100



Estimation observations: 107,916

Ruleout = 10

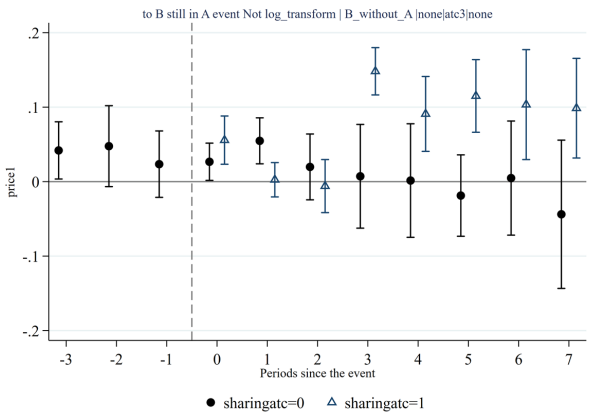


Estimation observations: 104,916

Notes. Black circles report estimates for `sharingatc=0`; hollow blue triangles report estimates for `sharingatc=1`. Error bars are 95% confidence intervals. The panels differ only in whether ruleout is off or in the price1-only threshold, and therefore in the resulting estimation sample. Dynamic sharing is disabled in all four runs.

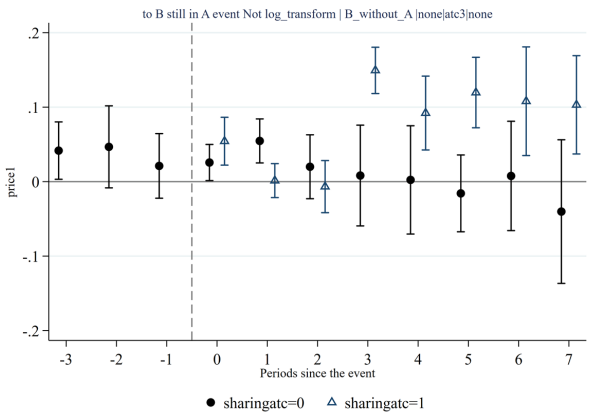
5 To B, Still in A — B-side Treatment

Ruleout = 0



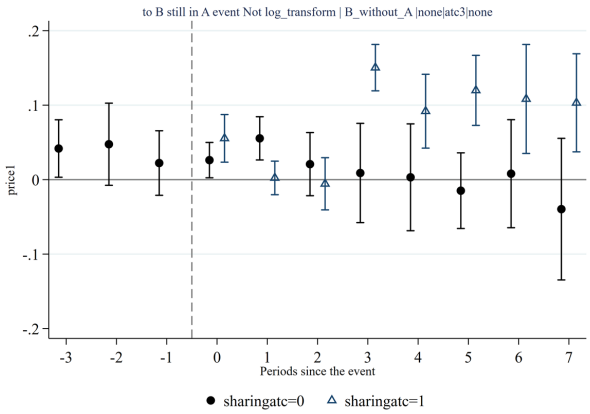
Estimation observations: 130,236

Ruleout = 1000



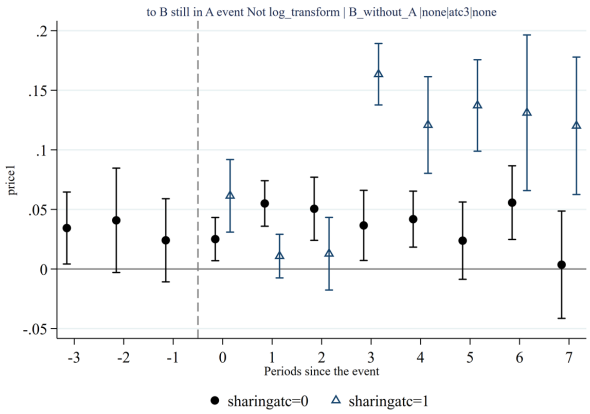
Estimation observations: 129,972

Ruleout = 100



Estimation observations: 129,480

Ruleout = 10

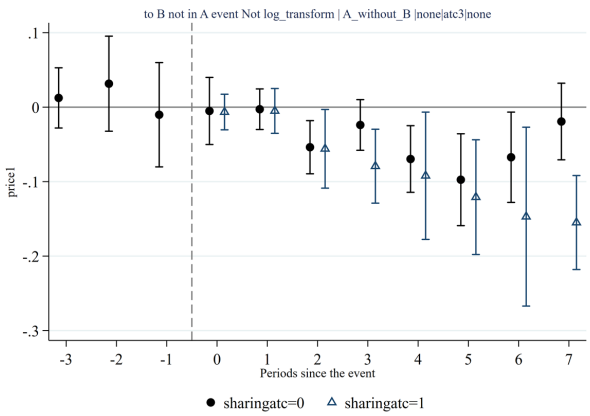


Estimation observations: 119,436

Notes. Black circles report estimates for `sharingatc=0`; hollow blue triangles report estimates for `sharingatc=1`. Error bars are 95% confidence intervals. The panels differ only in whether ruleout is off or in the price1-only threshold, and therefore in the resulting estimation sample. Dynamic sharing is disabled in all four runs.

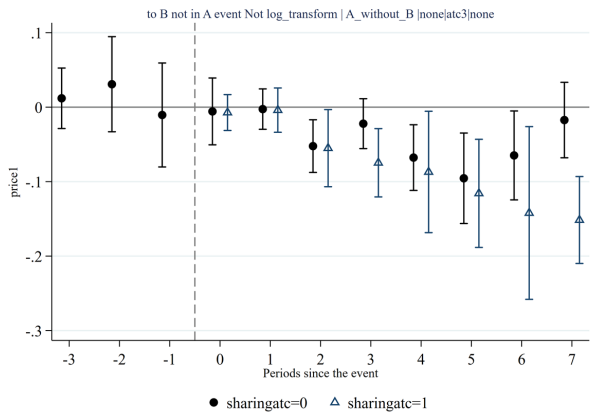
6 To B, Not in A — A-side Treatment

Ruleout = 0



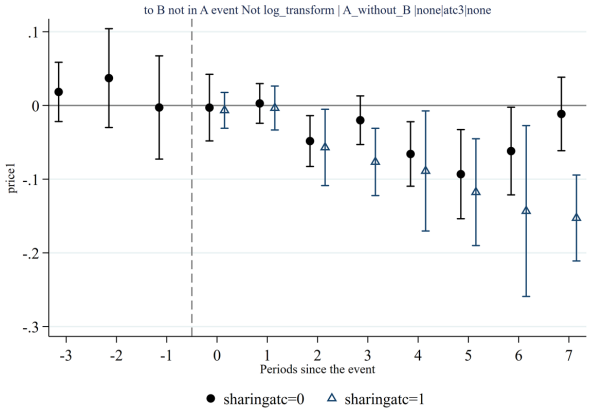
Estimation observations: 73,992

Ruleout = 1000



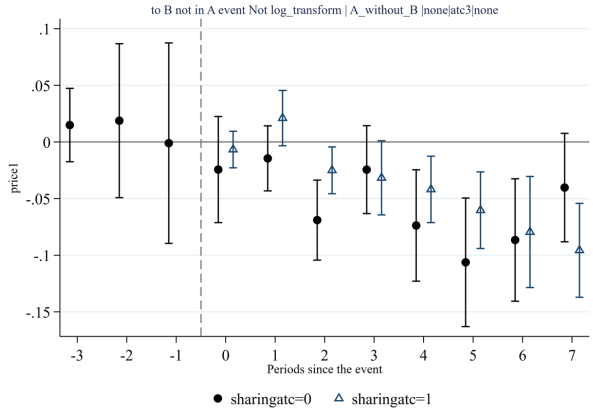
Estimation observations: 73,824

Ruleout = 100



Estimation observations: 73,536

Ruleout = 10

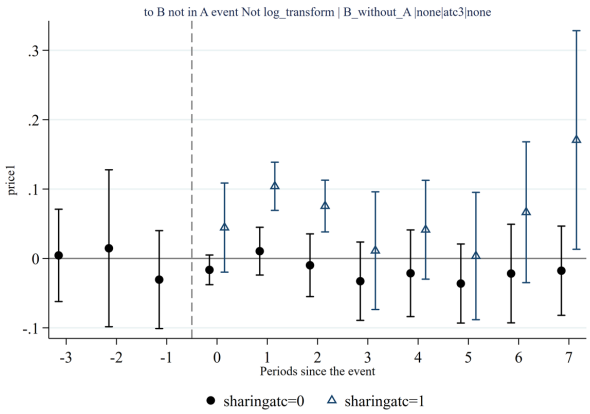


Estimation observations: 70,416

Notes. Black circles report estimates for `sharingatc=0`; hollow blue triangles report estimates for `sharingatc=1`. Error bars are 95% confidence intervals. The panels differ only in whether ruleout is off or in the price1-only threshold, and therefore in the resulting estimation sample. Dynamic sharing is disabled in all four runs.

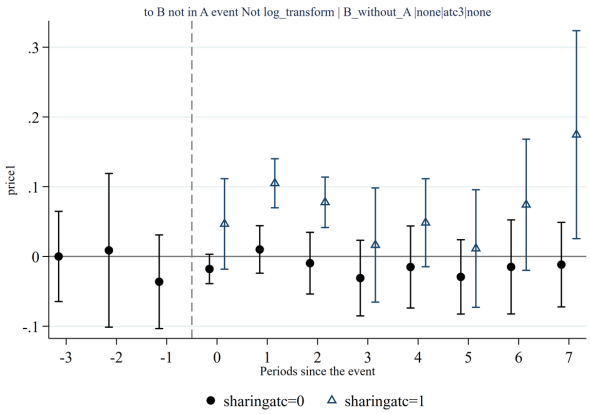
7 To B, Not in A — B-side Treatment

Ruleout = 0



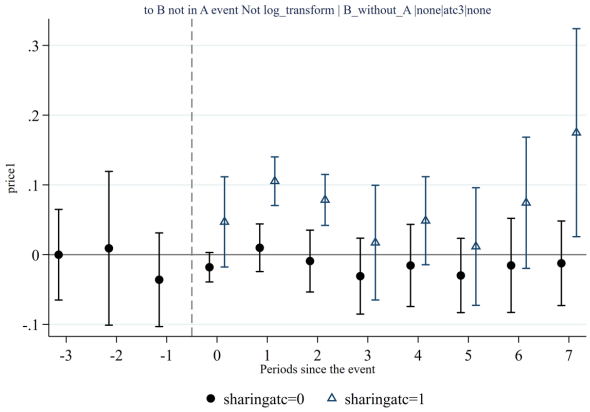
Estimation observations: 66,192

Ruleout = 1000



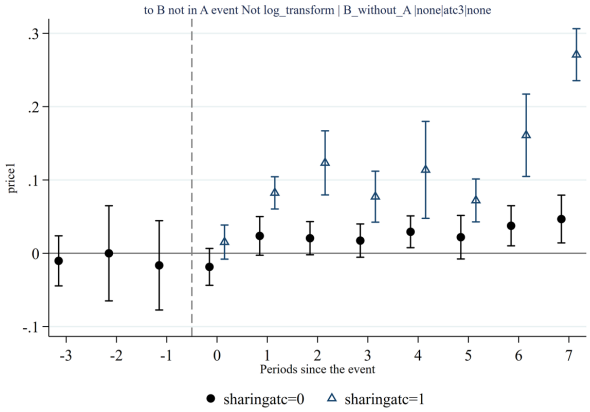
Estimation observations: 66,060

Ruleout = 100



Estimation observations: 65,952

Ruleout = 10



Estimation observations: 62,508

Notes. Black circles report estimates for `sharingatc=0`; hollow blue triangles report estimates for `sharingatc=1`. Error bars are 95% confidence intervals. The panels differ only in whether ruleout is off or in the price1-only threshold, and therefore in the resulting estimation sample. Dynamic sharing is disabled in all four runs.